

TIØ4145 Corporate Finance

Exercise 2: Valuation of stocks, capital markets and pricing of risk

Deadline: Monday September 8th, 2025, 12:00 PM – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday September 4th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on September 1th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapters 9 and 10

Tasks

Chapter 9 – Valuing Stocks

1. DFB, Inc., expects earnings at the end of this year of \$4.19 per share, and it plans to pay a \$2.43 dividend at that time. DFB will retain \$1.76 per share of its earnings to reinvest in new projects with an expected return of 15% per year. Suppose DFB will maintain the same dividend payout rate, retention rate, and return on new investments in the future and will not change its number of outstanding shares.
 - a. What growth rate of earnings would you forecast for DFB?
 - b. If DFB's equity cost of capital is 12.2%, what price would you estimate for DFB's stock today?
 - c. Suppose DFB instead paid a dividend of \$3.43 per share at the end of this year and retained only \$0.76 per share in earnings. If DFB maintains this higher payout rate in the future, what stock price would you estimate now? Should DFB raise its dividend?

2.  Heavy Metal Corporation is expected to generate the following free cash flows over the next five years:

Year	1	2	3	4	5
FCF (\$ millions)	52.2	68.7	77.2	75.6	80.5

After then, the free cash flows are expected to grow at the industry average of 4.1% per year. Using the discounted free cash flow model and a weighted average cost of capital of 14.9%:

- Estimate the enterprise value of Heavy Metal.
 - If Heavy Metal has no excess cash, debt of \$306 million, and 42 million shares outstanding, estimate its share price.
3. You notice that PepsiCo has a stock price of \$52.66 and EPS of \$3.20. Its competitor, the Coca-Cola Company, has EPS of \$2.49. Estimate the value of a share of Coca-Cola stock using only this data.
4.  Suppose that in January 2006, Kenneth Cole Productions had sales of \$518 million, EBITDA of \$55.6 million, excess cash of \$100 million, \$3 million of debt, and 21 million shares outstanding.

Ticker	Name	Stock Price (\$)	Market Capitalization (\$ millions)	Enterprise Value (\$ millions)	P/E	Price/Book	Enterprise Value/Sales	Enterprise Value/EBITDA
KCP	Kenneth Cole Productions	26.75	562	465	16.21	2.22	0.90	8.36
NKE	NIKE, Inc.	84.20	21,830	20,518	16.64	3.59	1.43	8.75
PMMAY	Puma AG	312.05	5,088	4,593	14.99	5.02	2.19	9.02
RBK	Reebok International	58.72	3,514	3,451	14.91	2.41	0.90	8.58
WWW	Wolverine World Wide	22.10	1,257	1,253	17.42	2.71	1.20	9.53
BWS	Brown Shoe Company	43.36	800	1,019	22.62	1.91	0.47	9.09
SKX	Skechers U.S.A.	17.09	683	614	17.63	2.02	0.62	6.88
SRR	Stride Rite Corp.	13.70	497	524	20.72	1.87	0.89	9.28
DECK	Deckers Outdoor Corp.	30.05	373	367	13.32	2.29	1.48	7.44
WEYS	Weyco Group	19.90	230	226	11.97	1.75	1.06	6.66
RCKY	Rocky Shoes & Boots	19.96	106	232	8.66	1.12	0.92	7.55
DFZ	R.G. Barry Corp.	6.83	68	92	9.20	8.11	0.87	10.75
BOOT	LaCrosse Footwear	10.40	62	75	12.09	1.28	0.76	8.30
Average (excl. KCP)					15.01	2.84	1.06	8.49
Max (relative to Avg.)					+51%	+186%	+106%	+27%
Min (relative to Avg.)					-42%	-61%	-56%	-22%

- Using the average enterprise value to sales multiple in the table above, estimate KCP's share price.
- What range of share prices do you estimate based on the highest and lowest enterprise value to sales multiples in the table?

- c. Using the average enterprise value to EBITDA multiple in the table, estimate KCP's share price.
- d. What range of share prices do you estimate based on the highest and lowest enterprise value to EBITDA multiples in the table?

Exam Question – December 2015

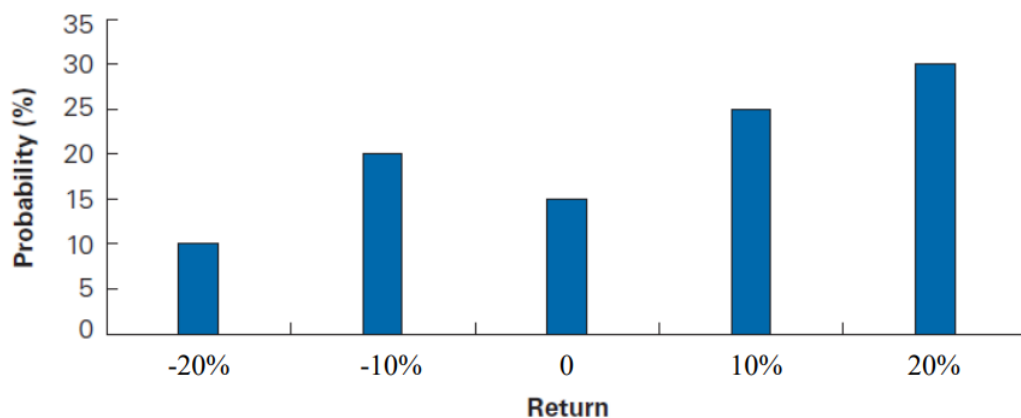
Stock A is expected to pay a dividend of \$15 next year. Thereafter, the dividend growth is expected to be 2% a year forever.

Stock B is expected to pay a dividend of \$10 next year. Thereafter, dividend growth is expected to be 4% a year for five years (i.e. until year 6) and zero thereafter.

If the equity cost of capital for both stocks is 12%, which stock is the most valuable?

Chapter 10 – Capital markets and the pricing of risk

1. The figure below shows the one-year return distribution for RCS stock. Calculate
 - a. The expected return
 - b. The standard deviation of the return



2. The last four years of returns for a stock are as follows:

Year	1	2	3	4
Return	-4%	28%	12%	4%

- a. What is the average annual return?
- b. What is the variance of the stock's returns?
- c. What is the standard deviation of the stock's returns?

3. Consider an investment with the following returns over four years:

Year	1	2	3	4
Return	6%	13%	-5%	13%

- What is the compound annual growth rate (CAGR) for this investment over the four years?
 - What is the average annual return of the investment over the four years?
 - What is a better measure of the investment's past performance?
 - If the investment's returns are independent and identically distributed, which is a better measure of the investment's expected return next year?
4. Consider two local banks. Bank A has 100 loans outstanding, each for \$1 million, that it expects will be repaid today. Each loan has a 5% probability of default, in which case the bank is not repaid anything. The chance of default is independent across all the loans. Bank B has only one loan of \$100 million outstanding, which it also expects will be repaid today. It also has a 5% probability of not being repaid. To understand the difference between the type of risk each bank faces, calculate:
- The expected overall payoff of each bank.
 - The standard deviation of the overall payoff of each bank.
 - Which bank faces less risk?
5. Consider an economy with two types of firms, S and I. S firms all move together. I firms move independently. For both types of firms, there is a 60% probability that the firms will have a 15% return and a 40% probability that the firms will have a -10% return. What is the volatility (standard deviation) of a portfolio that consists of an equal investment in 20 firms of (a) type S, and (b) type I?
6. Suppose the market portfolio is equally likely to increase by 30% or decrease by 10%.
- Calculate the beta of a firm that goes up on average by 43% when the market goes up and goes down by 17% when the market goes down.
 - Calculate the beta of a firm that goes up on average by 18% when the market goes down and goes down by 22% when the market goes up.
 - Calculate the beta of a firm that is expected to go up by 4% independently of the market.
7. Use the data from task 6. Suppose the risk-free interest rate is 4%.
- i. Use the beta you calculated for the stock in (6.a) to estimate its cost of capital

- ii. How does this compare to the stock's expected return?
- b.
 - i. Use the beta you calculated for the stock in (6.b) to estimate its cost of capital
 - ii. How does this compare to the stock's expected return?

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.